Kyungwon Kim, Ph.D.

Senior Research Scientist · Medical AI & Surgical Robotics

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SUMMARY

I received my Ph.D. in Electrical and Electronic Engineering from Yonsei University, where I worked on multimodal learning for cancer prognosis using medical imaging, genomics, and clinical data.

I am currently a research scientist at the Clinical Robotics and Embodied AI Research (CREAI) Center at Samsung Medical Center (SMC). My research is expanding toward surgical robotics, Physical AI, and world models, with a focus on developing intelligent systems that can perceive, reason about, and interact with complex clinical environments.

More broadly, my research interests span medical artificial intelligence, multimodal learning, computer vision, surgical robotics, and embodied intelligence.

PROFESSIONAL EXPERIENCE

Samsung Medical Center — Clinical Robotics and Embodied AI Research (CREAI) Center

Seoul, Korea

Senior Research Scientist — *Sep. 2026 - Present*

- Developing autonomous surgical robots; Vision-Language-Action (VLA) models; world models.

Genoray — AI Team, Imaging Group, FLEXLAB

Seoul, Korea

Senior Research Engineer — *Jul. 2026 - Sep. 2026*

- Developed AI-based methods to enhance X-ray image quality.

EDUCATION

Ph.D. in Electrical and Electronic Engineering

Seoul, Korea

Yonsei University (Advisor: Prof. Dosik Hwang, MAI-Lab) — *Mar. 2019 - Aug. 2026*

B.S. in Electrical and Electronic Engineering

Seoul, Korea

Yonsei University — *Mar. 2012 - Feb. 2019*

RESEARCH AREAS

Multi-modal Cancer Prognosis Prediction

MEDICAL

- Survival prediction for lung cancer patients using CT images, pathological images, and clinical data.
- Prognosis prediction for multiple cancer types using pathological images and genomic data under missing-modality settings.

Segmentation and Classification in X-ray and CT Images

MEDICAL

- Methods for automatic detection of anatomical or pathological regions in radiological images and disease classification based on image features.

Self-supervised Denoising of Low-dose X-ray Images

MEDICAL

- Deep learning–based methods for denoising low-dose X-ray images to approximate high-dose image quality without paired supervision.

Clinical Report Generation from 3D CT

MEDICAL

- Large Language Model (LLM)–based approaches for generating clinical reports, including findings and impressions, from 3D CT images.

Deep-Learning-based Fourier Ptychographic Microscopy (FPM)

OPTICS

- Techniques for reconstructing high-resolution optical images with a large field of view from multiple low-resolution images acquired under physical constraints.

PUBLICATIONS

MUST: Modality-Specific Representation-Aware Transformer for Diffusion-Enhanced Survival Prediction with Missing Modality — CVPR 2026

***Kyungwon Kim**, and Dosik Hwang* — *Jun. 2026, Denver, CO, USA*

h5-index 450 · arXiv Code

LLM-guided Multi-modal Multiple Instance Learning for 5-year Overall Survival Prediction of Lung Cancer — MICCAI 2024

***Kyungwon Kim**, Yongmoon Lee, Doohyun Park, Taejoon Eo, Daemyung Youn, Hyesang Lee, and Dosik Hwang* — *Oct. 2024, Marrakesh, Morocco*

FWCI 5.09 · Paper Code

M3F: Multi-Field-of-View Feature Fusion Network for Aortic Vessel Tree Segmentation in CT Angiography — MICCAI 2023

*Yunsu Byeon, Hyeseong Kim, **Kyungwon Kim**, Doohyun Park, Euijoon Choi, and Dosik Hwang* — *Oct. 2023, Vancouver, Canada*

MICCAI Challenge on Segmentation of the Aorta · Paper

Deep residual network with data consistency for subsampled Fourier Ptychographic Microscopy — SPIE Photonics West

*Hyeongyu Kim, **Kyungwon Kim**, Kyungchul Lee, Taejoon Eo, Seungah Lee, and Dosik Hwang* — *Jan. 2022, San Francisco, USA*

Paper

Are Two MR Images Enough to Generate the Third One Accurately? – Clinically Feasible Fat Suppression of Lumbar Spine MRI from T1w and T2w Images Only — ISMRM & SMRT

*Sewon Kim, Hanbyol Jang, **Kyungwon Kim**, Hyeon Gyu Kim, Young Han Lee, Sungjun Kim, and Dosik Hwang* — *Aug. 2020, Virtual*

Abstract

PATENTS

2026 — Method and Apparatus for Multimodal Data Analysis Supporting Missing Modalities Using Algebraic Representation Decomposition and Diffusion-Based Reconstruction, Application, KR 10-2026-0075495 — Korea

2022 — Method, Device and Recording Medium for Obtaining Parameter Data on an Object, Application, KR 10-2022-0039082 — Korea

RESEARCH EXPERIENCE

Segmentation of Oral Cancer and Lymph Node Metastasis from Head CT — Seoul, Korea

Finance All Solutions Co. · Project Manager — *Nov. 2025 – Aug. 2026*

- Developing deep learning–based methods to segment primary oral cancer lesions and metastatic lymph nodes from head CT scans collected from multiple institutions.

Detection and Segmentation of Osteomyelitis of the Jaw from 3D Dental CT — Seoul, Korea

Finance All Solutions Co. · Project Manager — *Jul. 2024 – Jun. 2025*

- Classifying the presence of osteomyelitis and identifying its precise location using 3D oral CT scans collected from six institutions.

Landmark Detection in Craniofacial 3D CT Data of Congenital Anomalies — Seoul, Korea

Finance All Solutions Co. · Project Manager — *Jul. 2024 – Jun. 2025*

- Detection of 45 landmarks in 3D craniofacial CT data for identifying congenital anomalies.

Report Generation from 3D CT using LLM — Montréal, Canada

Mila × Probe Medical — *Jun. 2024 – Sep. 2024*

- Participated in the 'Seoul AI Hub × Mila Partnership' as a researcher affiliated with Probe Medical.
- Development of a clinical report generating LLM from 3D CT.

Prognosis Prediction of Lung Cancer via Multi-modality — Seoul, Korea

Yonsei University × Probe Medical — *Sep. 2023 – Oct. 2024*

- Government-funded AI Dataset Construction Project (NIA).
- Development of a 5-year overall survival prediction model using 3D CT, pathological images, and clinical data.

Deep-learning-based Fourier Ptychographic Microscopy

Seoul, Korea

Yonsei University — *Sep. 2020 – Jan. 2022*

- Collaboration with the Optical Imaging Systems Laboratory (OISL), Yonsei University.

Denoising in X-ray Images

Seoul, Korea

Rayence Inc. — *Jun. 2021 – Dec. 2022*

- Development of a deep learning model that denoises and enhances the quality of low-dose X-ray images, helping reduce patients' X-ray exposure.

Disease Classification and Segmentation for Pediatric Chest X-ray

Seoul, Korea

Finance All Solutions Co. — *Jul. 2021 – Dec. 2021*

- Development of a deep learning model that classifies specific diseases and localizes lesions in pediatric chest X-ray images; establishment of a pediatric chest X-ray dataset.

Automatic Bone Segmentation in Cervical/Thoracic/Lumbar Spine Images

Seoul, Korea

Rayence Inc. — *Sep. 2019 – Sep. 2020*

- Development of an algorithm that automatically finds the spine in C/T/L spine X-ray images and calculates spinal curvature.

Object Area Detection in X-ray Images

Seoul, Korea

Rayence Inc. — *Sep. 2019 – Mar. 2020*

- Development of a deep learning model that automatically extracts only the object area, excluding the background, from medical X-ray images.
- Deployed in a commercial product.

Navigation Implant

Seoul, Korea

Osstem Implant Co. — *Jan. 2019 – Jun. 2020*

- Development of an algorithm to find the optimal implant location using tooth scan data consisting of 3D STL files.

Collimator Area Detection in X-ray Images

Seoul, Korea

Rayence Inc. — *Sep. 2018 – Mar. 2019*

- Developed an algorithm that automatically removes the collimator area, which is unnecessary information in X-ray images.
- Deployed in a commercial product.

TEACHING EXPERIENCE

Teaching Assistant for Prof. Dosik Hwang, Department of Electrical and Electronic Engineering, Yonsei University

Advanced Topics in Biomedical Engineering DELIVERED LECTURES — Mar. 2026 – Jun. 2026

Deep Learning Lab DELIVERED LECTURES — Mar. 2022 – Jun. 2022

Introductory Digital Labs — Sep. 2021 – Dec. 2021

Biomedical Engineering Experiments DELIVERED LECTURES — Sep. 2020 – Dec. 2020

Digital Signal Processing — Mar. 2020 – Jun. 2020

Biomedical Engineering Experiments DELIVERED LECTURES — Sep. 2019 – Dec. 2019

AWARDS

2024 Challenge Award, 2024 Yonsei Medical Center Big Data Challenge — Seoul, Korea

2023 3rd Place, MICCAI 2023 Grand Challenge SEG.A. — Segmentation of the Aortic Vessel Tree — Vancouver, Canada

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